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Principal Component Analysis (PCA)

considers a population, p -dimensional

$$\rightarrow \mu, \Sigma_{p \times p}$$

say we know μ, Σ for now

$P = [e_1, \dots, e_p]$ eigen values of Σ

$\Lambda = \text{diag}[\lambda_1, \dots, \lambda_p]$ eigen values

$$P^T P = I = P P^T \text{ (ortho-normal)}$$

$$\Sigma = P \Lambda P^T$$

$\downarrow$
sym pd

$\downarrow$
eigen val are real

in PCA theory we assume
 $\lambda_1 > \lambda_2 > \dots > \lambda_p > 0$

Although same λ 's may be repeated
but still Σ is always diagonalizable

Algebraic multiplicity of $\lambda =$ its geometric multiplicity

If I have multiple eigen vec for a λ
then P matrix may not be unique.

define $y_j = e_j^T x$ $1 \leq j \leq p$
 $1 \times p \quad p \times 1$

$$y = \begin{bmatrix} y_1 \\ \vdots \\ y_p \end{bmatrix} \quad p \times 1 \text{ random vector}$$

$$\begin{aligned} \text{Cov}(y_j, y_k) &= \text{Cov}(e_j^T x, e_k^T x) = e_j^T \Sigma e_k \quad (j \neq k) \\ &= \lambda_k e_j^T e_k = 0 \end{aligned}$$

(orthonormality)

$$\text{Var}(y_j) = \text{Cov}(y_j, y_j) = \lambda_j e_j^T e_j$$

$$\boxed{\text{Var}(e_j^T x) = \lambda_j = \text{Var } y_j \quad (1 \leq j \leq p)}$$

larger the eigen value the more imp it is
that is why we have arranged $\lambda_1 > \dots > \lambda_p$

But this var may come from outlier also

PCA tries to find principal components
first component it found is vector $a \in \mathbb{R}^p$ such that 'a' solves
 $\max a^T \Sigma a = \text{var}(a^T x)$

$\swarrow$
maximize vital info spread.

this optimization problem is unbounded from above

$$\begin{aligned} \max a^T \Sigma a \\ \text{st } \|a\|_2^2 = a^T a = 1 \end{aligned}$$

$\swarrow$
L-2 norm eq

$$\text{i.e. } \max a^T P \Lambda P^T a \quad \text{st } a^T a = 1$$

$$b = P^T a$$

$$\left(a^T a = a^T \underbrace{P P^T}_I a \right)$$

$$\max b^T \Lambda b \quad \text{ST} \quad \bar{b}^T b = 1 \quad \left(\quad = b^T \bar{b} = 1 \right)$$

$$\max \lambda_1 b_1^2 + \dots + \lambda_p b_p^2 \quad \text{ST} \quad b_1^2 + \dots + b_p^2 = 1$$

$$\lambda_1 b_1^2 + \dots + \lambda_p b_p^2 \leq \lambda_1 (b_1^2 + \dots + b_p^2) = \lambda_1$$

can be attained if $b_1 = 1$ and rest $b_i = 0$ ^{supremum}

$$b = P^T a$$

$$Pb = PP^T a = a$$

$$\therefore a = Pb = \begin{bmatrix} e_1 & \dots & e_p \end{bmatrix}_{p \times p} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{p \times 1}$$

$$= e_1 \quad (1^{\text{st}} \text{ PC})$$

$$\text{now } y_j = e_j^T x$$

$$y_1 = e_1^T x = a^T x$$

$$\text{var}(y_1) = \lambda_1$$

Second principal component is say finding a vector

$\hat{a} \in \mathbb{R}^p \setminus \{0\}$ such that

$$\max \hat{a}^T \Sigma \hat{a}$$

$$\text{st } \hat{a}^T \hat{a} = 1, \quad \text{cov}(\hat{a}^T x, \bar{x}) = 0$$

$$\hookrightarrow \hat{a}^T \Sigma \hat{a} = 0$$

$$\max \lambda_1 \hat{b}_1^2 + \lambda_2 \hat{b}_2^2 + \dots + \lambda_p \hat{b}_p^2$$

$$\text{st } \hat{b}_1^2 + \dots + \hat{b}_p^2 = 1$$

$$\hat{b} = P^T \hat{a} \Rightarrow \hat{a} = P \hat{b}$$

$$\hookrightarrow a^T \Sigma \hat{a} = 0$$

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